

The Title of the Article*

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Abstract

Here comes the abstract of the article, which is a summary of the main results discussed in the article. It should contain neither mathematical formulas, nor references.

Keywords: computer science, L^AT_EX, article format

1 Introduction

The paper should be written in standard L^AT_EX using the provided `actacyb.cls` class file. Please check for the latest version of the class file at the author kit GitHub page (https://github.com/attila10x/actacyb_author_kit).

Please do not use strange formats or non-standard packages. Avoid commands that would reconfigure the size of the page, the size and type of letters or the baselineskip.

The article in the first phase should be written including the `\showframe` package that makes the overfull boxes visible. Before sending the final version of your manuscript, please be sure that there are no such overfull boxes.

2 Using references

By default, Acta Cybernetica uses the `hyperref` package, making it easy to create links to various targets within the document. Hyperref displays these links in blue. These links are clickable and will navigate you to the target, either within or outside the document.

You can define a URL using the `\url` command. URLs are usually given in footnotes. For example:

*This work was supported by... If such a thanks text is included, use the `withtitlethanks` option of the `actacyb` document class!

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The Acta Cybernetica homepage¹ is the starting point for authors.

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Various elements in the document, such as figures, tables, sections, can be labelled using the `\label` command. For example, [Section 5](#) is defined as follows:

```
\section{Theorems and Lemmas}\label{sec:theorems}
```

These labels can be referenced from the text using either the commands `\ref` or `\autoref`. The former provides only the number of the element, while the latter also inserts the element's type as a string. For example:

```
You can read more about how to reference theorems and lemmas
in \autoref{sec:theorems}.
The \verb|\ref| command only provides the defined number,
in this case \ref{sec:theorems}.
```

You can read more about how to reference theorems and lemmas in [Section 5](#). The `\ref` command only provides the defined number, in this case [5](#).

You will find more examples of referencing in the following sections.

3 Pictures and figures

Every figure should appear in the `figure` environment, i.e. must be defined between `\begin{figure}` and `\end{figure}`. You should provide where you prefer to put your figure [`htbp`] (h – here, t – top, b – bottom, p – page), e.g., `\begin{figure}[!ht]` (insert figure here or top of next page if does not fit).

You can control the size of the included image in several ways as an option to `\includegraphics`. `scale` option defines the scaling factor of the image. `width` specifies the width of the image which can be combined with the value of `\linewidth`.

Primarily we use `pdflatex` to generate a PDF result of your document. As such, preferably include images in Portable Network Graphics (PNG) or Portable Document Format (PDF) formats with the e.g., `\includegraphics{example.png}` command. Alternatively $\LaTeX$ figures, PostScript(.ps) or Encapsulated PostScript (.eps) images can also be used with `latex` used for compiling.

Using drawing packages such as `tikz` or calling shell executables such as `pygmentize` may cause incompatibilities. If possible, try to generate a separate PDF file of these figures and include this PDF in the manuscript.

Before `\end{figure}` give a title to your figure within the `\caption{...}` command. To be able to refer to this figure later, label it with a unique identifier using the `\label{unique-label}` command. In the text you can refer to it with `\autoref{unique-label}`. Let us consider the following example:

¹<https://cyber.bibl.u-szeged.hu/index.php/actcybern/index>

```
An example can be seen in \autoref{fig:example}.
\begin{figure}[!ht]
\begin{center}
\includegraphics[width=.75\linewidth]{example.png}
\end{center}
\caption{The example figure\label{fig:example}}
\end{figure}
```

An example can be seen in [Figure 1](#).

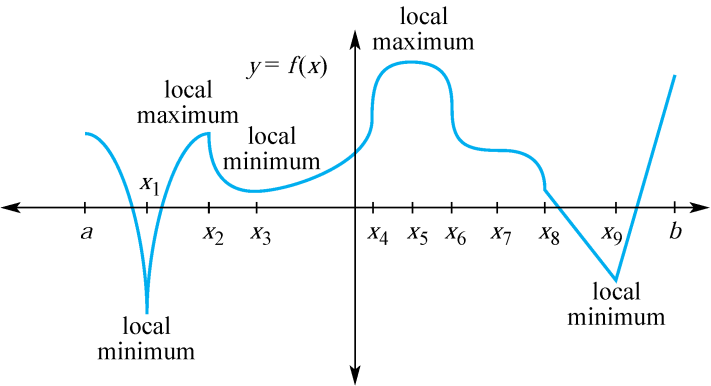


Figure 1: The example figure

4 Tables

Similarly to the procedure with figures, use the `table` environment to create tables. Normally, the title appears above the table, therefore put the `\caption{...}` command right after `\begin{table}[htbp]`. Use the `tabular` environment to prepare your table.

```
\begin{table}[!ht]
\caption{This is an example table\label{tab:res}}
\begin{center}
\begin{tabular}{l|rr}
Problem & Time(s) & Memory(MB) \\ \hline
easy    & 13      & 25 \\
hard    & 2359   & 6820
\end{tabular}
\end{center}
\end{table}
\autoref{tab:res} shows an example table.
```

Table 1: This is an example table

Problem	Time(s)	Memory(MB)
easy	13	25
hard	2359	6820

Table 1 shows an example table.

5 Theorems and Lemmas

In the `actacyb` class the `amsthm` package is included. Thus, the most common environments, such as `definition`, `theorem`, `lemma`, `proposition`, `corollary`, `claim`, `remark`, `example` and `proof` are already defined. To define other environments, please use the `\newtheorem{othersenv}[theorem]{othersenv}` command. **Please do not redefine the theoremstyle or theorem numbering!**

By default, each “theorem” type is consecutively numbered: **Theorem 1**, **Theorem 2**, **Lemma 1**, etc. throughout the paper. You may specify the selection of this behaviour explicitly in `\documentclass` at the beginning of the document as

```
\documentclass[theoremcontinuousnumbering]{actacyb}
```

If you need numbering by sections, you should indicate it as an option to `\documentclass` as

```
\documentclass[theoremsectionnumbering]{actacyb}
```

Not numbered theorem environments can also be used by specifying the option `notnumberedtheorems` for `documentclass`. With theorem numbering selected, it is also possible to use not numbered theorems using the asterisk (*) at the end of the environment name, e.g., `theorem*`, `definition*`, `remark*`.

The `proof` environment automatically generates the end of proof (qed) symbol at the end of the proof. If you don’t need this symbol, redefine it to an empty command inside the proof environment (for the given proof only) or in the preamble (globally) as `\renewcommand{\qedsymbol}{}`.

By using a theorem numbering option, the following commands produce the result below:

```
\begin{definition*}
We define a not numbered definition...
\end{definition*}

\begin{theorem}
If  $N=1$ , then  $P=NP$ .
```

```
\end{theorem}
```

```
\begin{proof}
```

It is easy to see that if $N=1$, then $NP=1 \cdot P = P$.

```
\end{proof}
```

```
\begin{remark*}
```

This is a not numbered remark.

```
\end{remark*}
```

```
\begin{definition}
```

We define....

```
\end{definition}
```

```
\begin{remark}
```

Let us note that the above theorem is not...

```
\end{remark}
```

Definition. *We define a not numbered definition...*

Theorem 1. *If $N = 1$, then $P = NP$.*

Proof. It is easy to see that if $N = 1$, then $NP = 1 \cdot P = P$. □

Remark. This is a not numbered remark.

Definition 1. *We define....*

Remark 1. Let us note that the above theorem is not...

By specifying the `theoremsectionnumbering` option at the `documentclass` declaration, we would see **Definition**, **Theorem 4.1**, **Remark**, **Definition 4.1**, and **Remark 4.1** captions here, respectively.

6 Equations

The usual way to write equations is to use the `equation` environment for a single line of equation, and the `eqnarray` environment for multiline equations. It is also possible to use the `array` environment within an `equation`, but we do not recommend its usage, because its spacing and labeling may not give the desired result. Besides the `equation` and `eqnarray` environments, other `amstex` environments, such as `multiline`, `align`, `split`, etc. are also recommended.

The function Φ is defined in `\autoref{eq:phi}`.

```
\begin{equation}
```

$$\Phi(x) = \prod_{i=1}^n \sin(x_i) \cos(x_i - \frac{1}{3}\pi) \label{eq:phi}$$

```
\end{equation}
```

The function Φ is defined in [Equation 1](#).

$$\Phi(x) = \prod_{i=1}^n \sin(x_i) \cos(x_i - \frac{1}{3}\pi) \quad (1)$$

7 Indentations

Below is an example of text indentation:

In addition, seeds have the following property:

```
\begin{itemize}
\item[]
If a seed of clause  $c_T$ , and example  $\{\mathbf{x}\}$  satisfies
 $c_T$  but not  $c$ , then  $\{\mathbf{x}\}$  has at least one attribute
in  $c_T$  that is not in  $c$ .\hfill(\tt*)
\end{itemize}
The procedure below...
```

In addition, seeds have the following property:

If a seed of clause c_T , and example $\mathbf{x}$ satisfies c_T but not c , then $\mathbf{x}$ has at least one attribute in c_T that is not in c . (*)

The procedure below...

7.1 Bulleted List

Below is an example of a bulleted list:

```
\begin{itemize}
\item for every  $x \in A$  and for...
\item for every  $x_1, x_2$  and for every...
\end{itemize}
```

- for every $x \in A$ and for...
- for every x_1, x_2 and for every...

7.2 Enumerated List

Below is an example of an enumerated list:

```
\begin{enumerate}
\item If one of the following conditions holds:
\begin{enumerate}
```

```

\item The first condition.\label{step:1a}
\item The second condition.
\end{enumerate}
\item If  $\frac{n}{n_1}=3$  then in the majority of cases the
assumption may be removed.
\end{enumerate}

```

Let us refer to the first condition by `\ref{step:1a}`.

1. If one of the following conditions holds:
 - a) The first condition.
 - b) The second condition.
2. If $\frac{n}{n_1} = 3$ then in the majority of cases the assumption may be removed.

Let us refer to the first condition by [1.a](#).

8 Algorithms

If you want to include algorithms in your manuscript, please use the `algorithm` package along with the `algorithmic` style. As in other floating environments, please use some of the `[htbp]` (h – here, t – top, b – bottom, p – page) letters right after `\begin{algorithm}` to determine the desired place of your algorithm. Please provide a title with the `\caption{...}` command, and use the `\label{alg:...}` and `\ref{alg:...}` commands if you want to refer to the algorithm in the text, as shown in the following example. If you have the latest `algorithmic` package, you can also refer to any step of the algorithm as shown in the following example.

A general Interval Branch and Bound algorithm is shown in Algorithm~`\ref{alg:ibb}`. One of the following selection rules is applied in Step `\ref{step:selrule}`:

```

\begin{algorithm}[ht!]
\caption{ A general interval B\&B algorithm.}
\label{alg:ibb}
\textbf{\underline{Funct}} IBB( $S, f$ )
\renewcommand{\algorithmiccomment}[1]{\hfill {\it #1}}
\begin{algorithmic}[1]
\STATE Set the working list  $\{\mathcal{L}\}_W := \{S\}$  and the final
list  $\{\mathcal{L}\}_Q := \{\}$ 
\WHILE{(  $\{\mathcal{L}\}_W \neq \emptyset$  )} \label{alg:igoend}
    \STATE Select an interval  $XX$  from  $\{\mathcal{L}\}_W$ 
        \label{step:selrule} \COMMENT{Selection rule}
    \STATE Compute  $\text{lb}(X)$  \COMMENT{Bounding rule}
    \IF[Elimination rule]{ $XX$  cannot be eliminated}

```

```

\STATE Divide  $X$  into  $X^j$ ,  $j=1, \dots, p$ , subintervals
\COMMENT{Division rule}
\FOR{ $j=1, \dots, p$ }
  \IF[Termination rule]{ $X^j$  satisfies the termination
\STATE Store  $X^j$  in  $\mathcal{L}_W$ 
\ELSE
  \STATE Store  $X^j$  in  $\mathcal{L}_W$ 
\ENDIF
\ENDFOR
\ENDIF
\ENDWHILE
\STATE \textbf{return}  $\mathcal{L}_Q$ 
\end{algorithmic}
\end{algorithm}

```

A general Interval Branch and Bound algorithm is shown in [Algorithm 1](#). One of the following selection rules is applied in [Step 3](#).

Algorithm 1 A general interval B&B algorithm.

Func $\text{IBB}(S, f)$

```

1: Set the working list  $\mathcal{L}_W := \{S\}$  and the final list  $\mathcal{L}_Q := \{\}$ 
2: while (  $\mathcal{L}_W \neq \emptyset$  ) do
3:   Select an interval  $X$  from  $\mathcal{L}_W$  Selection rule
4:   Compute  $lb f(X)$  Bounding rule
5:   if  $X$  cannot be eliminated then Elimination rule
6:     Divide  $X$  into  $X^j$ ,  $j = 1, \dots, p$ , subintervals Division rule
7:     for  $j = 1, \dots, p$  do
8:       if  $X^j$  satisfies the termination criterion then Termination rule
9:         Store  $X^j$  in  $\mathcal{L}_W$ 
10:      else
11:        Store  $X^j$  in  $\mathcal{L}_W$ 
12:      end if
13:    end for
14:  end if
15: end while
16: return  $\mathcal{L}_Q$ 

```

9 Bibliography

For the bibliography, it is mandatory to use `bibtex` to generate the items to achieve a consistent style! The `actaabbrv.bst` file defines the style descriptions that can be used like this:


```
\bibliographystyle{actaabbrv}
\bibliography{actabib}
```

Replace `actabib` with the name of your `.bib` file. Refer to an article [9], a book [5], a technical report [8], a manual [10], an article in a book [6], a chapter in a book [3], an unpublished paper [7], a thesis [2], a proceedings [4], or an article in proceedings [1] using the `\cite` command!

It is necessary to provide detailed description of the referred items. Make sure that only those items are included in the `bibtex` file that are actually cited from your manuscript! Please look up and provide the DOI whenever possible! If not available, provide an URL or ISBN. BibTeX generates clickable DOI, URL and ISBN links. Check that they work correctly!

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